

Roots of Resilience

Developer Implementation Prompts

Complete handoff prompts for restructuring the interactive dataviz story

Purpose. This document reproduces the complete set of prompts for the developer/computer-science collaborator, in the recommended order. The current dashboard should be evolved rather than rebuilt, with the existing visual identity and interactions preserved while the narrative is restructured around uncertainty, diversification, downside risk, and economic consequences.

Files to send alongside this PDF:

- premium_01_four_bets.svg/png
- premium_02_insurance_effect.svg/png
- premium_03_downside_exact_events.svg/png
- premium_04_winner_heatmap.svg/png
- premium_05_portfolio_frontier.svg/png
- premium_06_bootstrap_clean.svg/png
- viz_final_four_bets.csv
- viz_final_insurance_arrows.csv
- viz_final_downside_events.csv
- viz_final_downside_summary.csv
- viz_final_winner_heatmap.csv
- viz_final_portfolio_landscape.csv
- viz_final_pareto_frontier.csv
- viz_final_bootstrap.csv
- viz_final_joint_failure_summary.csv

Important: The SVG/PNG files are design references. The CSV files are the authoritative graph data for the interactive implementation.

PROMPT 1 - Architecture + non-negotiables

I am giving you the current Roots of Resilience GitHub Pages project.

Please work FROM THE EXISTING index.html. Do not redesign or rewrite the dashboard from scratch.

The existing site already has a strong visual identity and working interaction architecture that I want preserved:

- Ocean Garden / Field Notes theme toggle
- existing CSS variables and Pacific colour palette
- atmospheric background
- grain texture
- top bar
- progress rail
- scroll-driven story beats
- sticky visual card
- hectare allocation sliders
- garden plant-cell visual
- Tonga 1987 first stress test
- personalised 60-event backtest
- state.garden and hectareOutcome() logic
- Pacific map
- current photography
- responsive design
- reduced-motion behaviour
- accessibility and keyboard support

Current important IDs in index.html are:

hero
warming
plant
tonga
sixty
safe
fail
why
evidence
islands
resilience
meaning
close
methods

The current sticky visual slots are:

vs-warming
vs-garden
vs-tonga
vs-sixty
vs-safe
vs-fail
vs-why
vs-evidence
vs-islands

The current progress order is defined in:

```
const BEAT_IDS
```

and scene changes are controlled by:

```
const SCENE_FOR
```

The sticky visual label is controlled by:

```
const VISUAL_TITLES
```

```
showVisual(id)
```

The main personalised updates currently happen in:

```
updateAll()
renderGarden()
updateTonga()
updateSixty()
updateDecisionCard()
updateMixChip()
```

Do not break these existing working interactions.

NEW STORY OBJECTIVE

The dashboard should no longer feel like:

climate -> garden -> several statistical charts -> culture

It should build one coherent argument:

1. This matters to real people.
2. Warming raises the stakes of dry conditions.
3. The viewer makes a crop decision BEFORE knowing the future.
4. That decision faces one historical stress event.
5. Then the same decision faces 60 stress events.
6. The crop that performs best keeps changing.
7. The crops usually do not all fall below trend together.
8. Therefore diversification is not about predicting the winner.
9. It is about reducing exposure to choosing the wrong crop.
10. That difference can translate into meaningful farm-gate value.
11. Across the small price-covered sample, diversified exposure had substantially lower average downside.
12. But diversification did NOT win every year and this is NOT a universal crop recipe.
13. The appropriate crop mix remains local, cultural and contextual.
14. End with a call for local field testing rather than a universal prescription.

The final main story order should be:

```
hero
warming
plant
tonga
sixty
winner
joint
downside
money
insurance
islands
resilience
meaning
close
```

Update BEAT_IDS accordingly:

```
const BEAT_IDS = [
  'hero',
  'warming',
  'plant',
  'tonga',
  'sixty',
```

```
'winner',  
'joint',  
'downside',  
'money',  
'insurance',  
'islands',  
'resilience',  
'meaning',  
'close'  
];
```

Recommended SCENE_FOR:

```
const SCENE_FOR = {  
  hero:'hero',  
  warming:'warming',  
  plant:'plant',  
  tonga:'tense',  
  sixty:'tense',  
  winner:'tense',  
  joint:'clarity',  
  downside:'clarity',  
  money:'golden',  
  insurance:'clarity',  
  islands:'archipelago',  
  resilience:'clarity',  
  meaning:'golden',  
  close:'sunset'  
};
```

Replace the existing MAIN-STORY beats:

safe
fail
why
evidence

with:

winner
joint
downside
money
insurance

Do NOT destroy the underlying old correlation / optimiser /
sensitivity functionality. Move those into an optional
"Explore the model" section later.

Update the sticky visual system to include:

vs-winner
vs-joint
vs-downside
vs-money
vs-insurance

and retain:

vs-warming
vs-garden
vs-tonga
vs-sixty
vs-islands

Update VISUAL_TITLES to something like:

```
const VISUAL_TITLES = {  
  warming:'Warming',
```

```
garden:'Your hectare',
tonga:'First stress test',
sixty:'60 stress tests',
winner:'The winner keeps changing',
joint:'56 of 60',
downside:'The downside tail',
money:'Fiji 2004 · farm-gate value',
insurance:'Economic downside',
islands:'Local context'
};
```

IMPORTANT:

The current desktop .visual-card max-width is only 520px.

The new winner heatmap, downside event plot and insurance chart need more horizontal room.

Add a wide visual-card state, for example:

```
.visual-card.wide {
  max-width: 760px;
}
```

and inside showVisual(id), toggle it for:

```
winner
downside
money
insurance
```

Do not simply increase every card permanently because the garden, Tonga and map look better at their current compact size.

Do not change any quantitative result while doing this architecture refactor.

First complete the structural refactor and make sure the existing warming, garden, Tonga, sixty-event backtest and map still work before building the new graphs.

PROMPT 2 - Data loading and integrity

I am also providing R-generated CSV files for the new visualisations.

Treat the R outputs as authoritative.

Do NOT recompute the economic analysis in JavaScript from scratch.

The browser is for VISUALISATION and INTERACTION, not for changing the statistical methodology.

Create a /data directory if needed and organise the supplied files cleanly.

Expected datasets include:

viz_final_four_bets.csv
viz_final_insurance_arrows.csv
viz_final_downside_events.csv
viz_final_downside_summary.csv
viz_final_winner_heatmap.csv
viz_final_portfolio_landscape.csv
viz_final_pareto_frontier.csv
viz_final_bootstrap.csv
viz_final_joint_failure_summary.csv

The existing STRESS_EVENTS JavaScript array can remain because the existing user-controlled garden simulation depends on it.

The new R-generated data should drive the fixed evidence graphics.

Use either:

A) the existing vanilla SVG approach, OR
B) D3 v7 for rendering/scales/data joins.

If using D3:

- use it only as a visualisation library
- do not use it to change methodology
- preferably vendor the minified D3 file locally rather than depend on a CDN
- keep the existing site lightweight

Build one async data-loading function such as:

```
loadEvidenceData()
```

and load all new datasets once at startup.

Show a tasteful loading state while they load.

If one dataset fails to load:

- do not leave a blank screen
- show an accessible fallback message
- keep the rest of the story functional

Do not hardcode displayed statistical numbers if they are present in the CSV.

Calculate display counts such as 7/10 or 25/60 directly from the loaded final visual data where practical.

However, do NOT mutate or transform the underlying figures into a different statistic.

Quantitative integrity checks that the final site must reproduce:

MAIN PHYSICAL SAMPLE

- 6 countries
- 10 selected unusually dry years per country
- 60 stress events total

BEST CROP COUNTS

- Sweet potato: 25 / 60
- Taro: 21 / 60
- Cassava: 14 / 60

WORST CROP COUNTS

- Sweet potato: 23 / 60
- Taro: 23 / 60
- Cassava: 14 / 60

JOINT PATTERN

- all three national crop-yield series below fitted trend: 4 / 60
- at least one at or above fitted trend: 56 / 60
- 4 / 60 = 6.67%, NOT 400%

CORE EQUAL-THIRDS PHYSICAL RESULT

- worst event approximately -36.7%
- best monocrop worst case approximately -48.9%
- worst-case loss magnitude reduction approximately 24.8%

PRICE-COVERED ECONOMIC SAMPLE

- 10 events only
- Fiji + New Caledonia only

ECONOMIC DOWNSIDE

- single-crop benchmark mean downside $\approx 6.63\%$
- diversified mean downside $\approx 3.10\%$
- proportional reduction $\approx 53.3\%$
- diversification reduced downside in 7 events
- increased downside in 2
- equal in 1

FIJI 2004

Cassava only:

expected $\approx$ US\$3,913
actual $\approx$ US\$4,833
change $\approx$ +US\$920
outcome $\approx$ +23.5%

Taro only:

expected $\approx$ US\$6,912
actual $\approx$ US\$5,944
change $\approx$ -US\$968
outcome $\approx$ -14.0%

Sweet potato only:

expected $\approx$ US\$1,930
actual $\approx$ US\$1,250
change $\approx$ -US\$680
outcome $\approx$ -35.3%

Equal-third diversified:

expected $\approx$ US\$4,251
actual $\approx$ US\$4,009
change $\approx$ -US\$243
outcome $\approx$ -5.7%

STATISTICAL SUPPORT

- exact paired sign-flip $p \approx 0.0078$
- paired Wilcoxon $p \approx 0.0122$
- event bootstrap 95% interval approximately 1.37 to 5.91 percentage points

Do not alter these values because the front-end should match the reproducible R analysis.

PROMPT 3 - Replace 'safe' with the interactive winner heatmap

Now implement the new "winner" beat.

Replace the current:

```
id="safe"  
data-visual="safe"
```

with:

```
id="winner"  
data-visual="winner"
```

Remove the current Best Performer / Worst Performer mini bar visual from the main story.

Use:

viz_final_winner_heatmap.csv

and use:

premium_04_winner_heatmap.svg

only as the visual DESIGN reference.

Do not embed the SVG as the final chart.
Rebuild the chart as an interactive SVG.

COPY

Eyebrow:

NO SAFE CROP

Headline:

The winner keeps changing.

Main copy:

Across 60 unusually dry country-years, no root stays on top.
Which crop you would have wanted depends on the place and the year —
information you only learn after the shock.

Then a prominent secondary line:

That is the problem with betting on one crop.

VISUAL

Create six country panels:

Fiji
New Caledonia
Niue
Papua New Guinea
Solomon Islands
Tonga

Each panel contains:

3 rows:
Cassava
Taro
Sweet potato

10 columns:
the country's ten selected dry stress years

Each tile should show crop yield anomaly relative to that crop's fitted trend.

Use a diverging colour scale centred exactly on zero:

negative = stressed / below trend
zero = neutral
positive = above trend

Use a perceptually sensible palette matching the supplied R reference.

The colour scale may be visually capped at $\pm 50\%$ for legibility,
BUT DO NOT ALTER THE UNDERLYING DATA.

Put a small star on the crop with the highest yield anomaly in each country-year.

INTERACTION

Hover / tap on one tile:

Highlight that tile and show:

Country
Year
Crop
Yield anomaly vs fitted trend
Rainfall anomaly if present
Temperature anomaly if present
"Best-performing root in this event" if applicable

The tooltip should never say:
"drought caused X% yield change"

This is descriptive historical co-occurrence, not causal attribution.

When the user hovers/taps a crop label such as Sweet potato:

- highlight all Sweet potato cells across all six country panels
- dim cassava and taro to around 20–30% opacity

This should let the viewer visually follow one crop through history.

When the user hovers one YEAR/event, if practical:
highlight all three crop cells for that same country-year.

ANNOTATION

Add small dynamic count chips:

Best performer:
Sweet potato 25
Taro 21
Cassava 14

Do not frame Sweet potato as "the best crop".

The point is precisely that the winner changes.

Caption:

60 selected unusually dry country-years — 10 per country.

Colour shows national crop-yield deviation from fitted trend.

Source: FAOSTAT crop data + Pacific Data Hub rainfall anomalies.

Mobile:

- maintain readable crop labels
- allow horizontal scrolling inside individual country panels if needed
- tap replaces hover
- do not shrink labels into illegibility

PROMPT 4 - Fix the 56/60 section and language

Now replace the current "fail" beat with a new "joint" beat.

Current problematic copy includes phrases such as:

"How often did all three roots fail together?"

"All three failed"

Do not use those phrases anymore.

The national data do not tell us that every root crop or every household garden literally failed.

Use:

id="joint"

data-visual="joint"

COPY

Eyebrow:

THE PATTERN THAT MAKES DIVERSIFICATION POSSIBLE

Headline:

They usually didn't all fall together.

Large number:

56 / 60

Main copy:

In 56 of the 60 selected unusually dry country-years, at least one of the three national crop-yield series remained at or above its fitted trend.

Secondary:

Only 4 / 60 had all three below fitted trend together.

Optional smaller clarification:

This is a national crop-series result — not a claim that every household garden followed the same pattern.

VISUAL

Keep the useful 60-tile concept from the existing fail visual but redesign the categories.

Green / positive category label:

At least one crop series at/above trend

Red / negative category label:

All three below fitted trend

NEVER label the red category:

"All three failed"

The 4 red events are:

Fiji 2006

New Caledonia 2014

Tonga 1980

Tonga 1981

Hover/tap on any tile:

Country

Year

Cassava anomaly

Taro anomaly

Sweet potato anomaly

For the 4 red events display:

"All three below fitted trend"

For other events:

"At least one at or above fitted trend"

Do not display the erroneous 400% value anywhere.

If displaying percentages:

$4/60 = 6.67\%$

$56/60 = 93.33\%$

PROMPT 5 - Build the 60-event downside graph

Now replace the current "why" main-story correlation beat with:

```
id="downside"  
data-visual="downside"
```

The existing correlation explorer should NOT be deleted permanently.
Move it later into "Explore the model".

Use:

```
viz_final_downside_events.csv  
viz_final_downside_summary.csv
```

Design reference:

```
premium_03_downside_exact_events.svg
```

COPY CHOREOGRAPHY

Eyebrow:

INSURANCE, NOT PREDICTION

Initial headline:

Diversification isn't trying to pick the winner.

As the visual comes into view, reveal a second line:

It's trying to make the loser hurt less.

Subcopy:

The same 60 dry stress events were applied to each strategy.
Look at how far the worst losses stretch.

VISUAL

Create four horizontal lanes:

Cassava only
Taro only
Sweet potato only
Mixed garden — 1/3 each

Every dot represents one of the SAME 60 country-year stress events.

Horizontal position:

$\text{downside} = \max(-\text{harvest outcome } \%, 0)$

A value of zero means that strategy did not finish below its fitted
harvest benchmark in that stress event.

Do not use a smoothed density/KDE.

There are many genuine zero observations and the raw event dots are
more truthful and more interpretable.

Use a swarm/beeswarm/dot-stack so repeated zeroes remain visible.

Draw a vertical dashed threshold at:

10%

Label it:

Severe-loss threshold

Keep the Mixed garden lane visually prominent using the existing gold accent.

Do not hide monocrop successes.

MOST IMPORTANT INTERACTION

Give every observation a stable country-year ID.

When the viewer hovers/taps ONE dot — for example Fiji 1992 under Taro only — highlight the corresponding Fiji 1992 observation across ALL FOUR strategy lanes.

Fade all other observations to around 20–25%.

This is crucial.

It should visually answer:

SAME HISTORICAL EVENT
→ DIFFERENT PLANTING STRATEGY
→ DIFFERENT DOWNSIDE

Tooltip:

Country · Year
Strategy
Harvest outcome %
Downside %
Rainfall anomaly if available

SUMMARY ANNOTATIONS

At the right of each lane, show compact statistics such as:

severe events / 60
worst decline

Do not make a false claim that the equal-third strategy has the fewest severe events of every strategy.

Sweet potato and the equal-third garden both have 7 severe-loss events under the current threshold.

The useful distinction is particularly in the lower-tail magnitude.

The equal-third garden's historical worst decline is about -36.7%, versus -48.9% for the best monocrop worst case.

Caption:

Conditional historical stress-test outcomes, not annual drought probabilities or causal yield effects.

PROMPT 6 - Build the Fiji 2004 economic reveal

Add a NEW main-story beat:

id="money"

data-visual="money"

Use:

viz_final_four_bets.csv

Design reference:

premium_01_four_bets.svg

COPY

Eyebrow:

WHAT DOES A BAD BET COST?

Headline:

One hectare. Same dry year. Four bets.

Subheadline:

Fiji · 2004

Short explanation:

The crop choice changes not just tonnes harvested, but the gross farm-gate value attached to that harvest.

VISUAL

Use four horizontal diverging bars centred at zero.

IMPORTANT:

BAR LENGTH MUST REPRESENT THE PERCENTAGE FARM-GATE-VALUE OUTCOME, NOT RAW DOLLARS.

This is necessary because the four strategies begin with different expected farm-gate values per hectare.

Place the dollar difference as a large text annotation on each bar.

Display:

Cassava only

+23.5%

+US\$920

Taro only

-14.0%

-US\$968

Sweet potato only

-35.3%

-US\$680

Mixed garden — 1/3 each

-5.7%

-US\$243

The visual should clearly show zero in the centre:

LEFT = below fitted benchmark

RIGHT = above fitted benchmark

Use the site's existing crop colours and gold for the diversified strategy.

HOVER / TAP

Each bar tooltip must show:

Strategy

Expected/fitted farm-gate value

Actual 2004 farm-gate value

Dollar difference

Percentage difference

Approximate values:

Cassava:

expected US\$3,913

actual US\$4,833

Taro:

expected US\$6,912

actual US\$5,944

Sweet potato:

expected US\$1,930

actual US\$1,250

Mixed:

expected US\$4,251

actual US\$4,009

STORY REVEAL

After the four bars animate in, reveal:

"If you knew cassava would win, you wouldn't need diversification."

Then reveal:

"But you only know the winner after the year has happened."

This line is central to the project's insurance argument.

Do NOT make the viewer conclude:

"Cassava is the climate-proof crop."

Cassava happened to perform best in this historical event.

TERMINOLOGY

Use:

gross farm-gate value
gross farm-gate value difference
gross value of harvest

Do NOT call the values:

profit
farmer income
household income
retail value
welfare loss

Also do not say Sweet potato had the largest dollar loss.

Sweet potato had the largest proportional loss (-35.3%).

Taro had the largest absolute dollar shortfall (-US\$968) because the expected value of the taro hectare was substantially larger.

Caption:

Observed same-year FAOSTAT producer prices are used to value the historical harvest quantities. This is not a model of drought-induced price changes.

PROMPT 7 - Build the 53.3% insurance result

Add a NEW main-story beat:

id="insurance"
data-visual="insurance"

Use:

viz_final_insurance_arrows.csv

Design reference:

premium_02_insurance_effect.svg

COPY

Eyebrow:

PRICE-COVERED STRESS TEST

Headline first:

Diversification didn't win every year.

Then reveal:

It made losing less costly.

Large statistic:

6.6% → 3.1%

Underneath:

53.3% lower average downside

Precise supporting sentence:

Across the 10 selected dry stress events with complete producer prices for all three crops, average farm-gate-value downside was 6.6% under an equal-probability single-crop benchmark and 3.1% for the equal-third diversified hectare.

Visible caveat directly underneath:

Price coverage is limited to 10 events in Fiji and New Caledonia.
Treat this economic result as exploratory.

VISUAL

Create one horizontal paired arrow/dumbbell per event.

10 rows total.

The events come from:

Fiji:
1998
2003
2004
2006

2015

New Caledonia:

2015

2016

2017

2018

2019

X axis:

Farm-gate-value downside %

LOWER / LEFT = less downside

Start point:

equal-probability single-crop benchmark

Endpoint:

equal-third mixed hectare

Use:

grey start circle = single-crop benchmark

gold endpoint = diversified hectare

Arrow/segment colour:

green =

diversification reduced downside

red =

diversification increased downside

neutral =

no meaningful difference

The current pattern is:

7 events reduced downside

2 increased downside

1 equal

DO NOT hide the red exceptions.

The chart is stronger because it shows that diversification did not automatically outperform in every year.

INTERACTION

Hover/tap one event:

Country

Year

Single-crop benchmark downside %

Diversified downside %

Difference in percentage points

Direction: reduced / increased / equal

Make the corresponding row stronger and fade the other rows.

If practical, provide a "Play all 10 events" animation that moves from top to bottom once, but:

- never loop continuously
- respect prefers-reduced-motion
- make the chart fully understandable without animation

EXPLAIN THE BENCHMARK

Add a small accessible info button:

"What is the single-crop benchmark?"

On click/tap explain:

"The benchmark asks what downside would be expected if the whole hectare had been committed to one of cassava, taro or sweet potato before the outcome was known, giving the three crop choices equal weight. It is a hypothetical ex-ante risk benchmark, not a description of actual farmer planting behaviour."

STATISTICAL SUPPORT

Do not make p-values the headline.

Place them in a small expandable "Statistical check" block:

Observed average downside benefit:
 ≈ 3.53 percentage points

Exact paired sign-flip:
 $p \approx 0.0078$

Paired Wilcoxon:
 $p \approx 0.0122$

Event bootstrap:
95% interval ≈ 1.37 to 5.91 percentage points

Then state:

"These tests are supporting exploratory evidence. The price-covered sample has only 10 events clustered within two countries, so the effect size and event-level pattern are more important than treating the p-values as definitive Pacific-wide inference."

VERY IMPORTANT:

Do NOT translate "53.3% lower downside" into
"53.3% fewer dollars lost".

The formal result concerns the percentage downside in farm-gate value.

Raw dollar shortfalls are used only for within-event illustrations such as Fiji 2004.

PROMPT 8 - Reposition the Pacific map

Keep the existing "islands" section and the existing Pacific map.

Do not remove the map.

Change its narrative role.

COPY

Eyebrow:

LOCAL, NOT UNIVERSAL

Headline:

The same mix does not hedge every island the same way.

Main copy:

The historical data support diversification as a risk-management principle. They do not prescribe one Pacific crop mix.

Secondary copy:

Different crops move together differently across countries, and national harvested-area patterns also differ.

INTERACTION

Keep the existing map markers for:

Fiji
New Caledonia
Niue
Papua New Guinea
Solomon Islands
Tonga

Keep country selection and its descriptive correlation information.

Keep the "effective number of selected roots" information, but label it precisely:

"balance among cassava, taro and sweet potato only"

Do not label it:

"total crop diversity"
"agricultural biodiversity"
"household garden diversity"

The existing COUNTRY_POS structure and map drawing logic can remain.

The existing focus pill can remain.

Use this scene as an intentional corrective immediately after the 53.3% result:

53.3% lower downside in the price-covered sample
DOES NOT mean
one-third cassava + one-third taro + one-third sweet potato is the

correct recipe for every Pacific community.

Add a small line:

"Principle ≠prescription."

PROMPT 9 - Move technical graphs into 'Explore the model'

Create a new optional section called:

Explore the model

Place it AFTER the main narrative close or make it accessible from a button near Methods.

It should not be part of BEAT_IDS and should not interrupt the main scroll story.

Use expandable cards/details or a separate exploration panel.

Move the old main-story analytical content here:

1. full country correlation explorer
2. strategy worst-case comparison
3. $\pm 50\%$ anomaly sensitivity toggle
4. optimiser results
5. 231-garden risk/return frontier
6. bootstrap evidence

PORTFOLIO FRONTIER

Use:

viz_final_portfolio_landscape.csv
viz_final_pareto_frontier.csv

Design reference:

premium_05_portfolio_frontier.svg

Every dot = one tested one-hectare crop allocation.

There are 231 allocations using 5-percentage-point steps.

X axis:

Worst historical stress-event outcome

Moving right =
less severe worst case

Y axis:

Mean historical stress-event outcome

Colour:
effective number of selected crops

Size:
number of $\geq 10\%$ severe-loss stress events

Highlight:

Equal thirds

and:

Best historical worst-case mix
10% cassava
40% taro

50% sweet potato
worst historical result $\approx$ -18.6%

Draw the historical Pareto frontier.

Hover one point:

Cassava %
Taro %
Sweet potato %
Mean outcome
Worst outcome
Severe-loss events

HIGH-VALUE INTERACTION:

When the user clicks a portfolio point, allow:

"Try this mix in my garden"

If clicked:
set state.garden to that crop allocation
update the three sliders
call updateAll()
scroll back to the sixty-event personalised backtest

This makes the technical explorer connect back to the user's original decision.

CRITICAL wording:

Never call the 10/40/50 mix:
"The optimal Pacific garden"

Call it:
"best historical worst-case mix in this sample"

Make the following caveat visible:

"In-sample historical optimisation. Not a planting recommendation."

OLD CORRELATIONS

Move the current updateCorr() visual here.

Label:

"Full-record national yield correlations"

Keep the current country buttons.

Explain:

"Correlation describes whether the detrended national yield series tend to move together. It is not a causal drought-response coefficient."

SENSITIVITY

Keep the existing $\pm 50\%$ anomaly cap toggle.

Explain what it does.

Use the current result to show that the broad lower-tail conclusion does not disappear when extreme crop anomalies are visually/statistically capped.

Do not present this as proof.

BOOTSTRAP

Use:

viz_final_bootstrap.csv

Design reference:

premium_06_bootstrap_clean.svg

Place it in this technical section rather than the main narrative.

Label the zero line clearly.

Show the 95% interval.

Explain that this is event-level bootstrap support for a small price-covered sample, not causal or Pacific-wide inference.

PROMPT 10 - Refine culture, knowledge and ending

Now refine the existing resilience / meaning / close sections.

Keep:

img-resilience.jpg

img-taro.jpg

img-hope.jpg

Keep the existing visual style and full-width emotional photography.

RESILIENCE IMAGE

The current image reveal says:

"In 56 of 60 dry years, at least one root stayed at or above trend."

Change this slightly to:

"In 56 of 60 selected dry country-years, at least one national crop series stayed at or above its fitted trend."

Keep the hover/tap unveiling interaction.

Caption:

"Different crops do not always struggle together."

Do not say this image itself depicts one of the measured events.
It is narrative imagery.

MEANING

Keep the headline:

"A garden can be statistically resilient and still fail its community"

Replace the current over-broad taro sentence.

Use:

"Yield is only one dimension of resilience. In Pacific communities, crop choices can also carry cultural, ceremonial, exchange, labour, taste and place-based significance that tonnes alone cannot capture."

If mentioning taro specifically, use qualified wording such as:

"In places including Samoa and Fiji, taro also carries important ceremonial and exchange value."

Do not write:

"taro alone carries ceremonial weight across the Pacific"

because that overgeneralises many different Pacific cultures.

KNOWLEDGE CARDS

In the current KNOWLEDGE JavaScript array, the Bellona card currently

contains:

"the same principle this dataset measures another way."

Replace that wording with:

"a broader resilience principle also visible in the historical data."

Suggested Bellona text:

"On Bellona, shifting cultivation with many crops and cultivars has formed part of local risk management — a broader resilience principle also visible in the historical data."

The dashboard must NOT imply:

- this model discovered diversification
- this model validated Indigenous knowledge
- Pacific farmers need outsiders to teach them crop diversity

Instead frame the quantitative work as:

"historical national records illustrating why a long-recognised resilience principle can matter."

CLOSE

Keep:

"There is no climate-proof crop."

Keep:

"There may not even be a climate-proof garden."

Then use:

"Diversity gives a garden more ways to absorb uncertainty."

Replace the current main funding line:

"Fund the ability to choose — not one prescription."

with the stronger climax:

"Fund the test before funding the prescription."

Then show, line by line:

Preserve the planting material.

Run the local trials.

Measure yield, labour, income, nutrition and food security.

Let communities determine what works where they are.

Keep the existing GCF and CePaCT funding examples but introduce them as examples of relevant climate-resilience infrastructure / investment.

Do not imply those projects have already validated the exact cassava+taro+sweet-potato portfolio analysed here.

Final CTA buttons can remain:

Try a different mix
Methods & sources

Also add:

Explore the model

PROMPT 11 - Replace and expand Methods precisely

Update the current Methods & sources section.

Keep the existing physical-analysis explanation but rewrite it cleanly into subsections.

PHYSICAL STRESS TEST

Use:

"Crop production and harvested area come from FAOSTAT, 1961–2024. Rainfall and surface-temperature anomalies come from the Pacific Data Hub Climate Change Indicators dataset, SPC:DF_CLIMATE_CHANGE(1.0).

For each of the six analysis countries, the selected stress sample is the lowest 20% of available rainfall-anomaly observations by count. With 46 rainfall observations from 1979–2024, this gives 10 selected dry years per country and 60 country-year stress events in total.

Crop yields are detrended separately by country and crop using a log-linear time trend.

The interactive hectare is a hypothetical land allocation. Expected and historical harvests weight each crop by its fitted tonnes-per-hectare yield and the user's land share. It is not a simple arithmetic average of crop percentage anomalies."

INTERPRETATION

Use:

"The stress test is descriptive, not causal. National crop statistics are not household-garden observations, and selected historical country-years are not forecasts of future farm outcomes."

Also:

"'Unusually dry' is the project's analytical definition — the lowest 20% of rainfall-anomaly observations by country — not an official severe-drought classification."

PRODUCER PRICE EXTENSION

Add:

"Producer-price data come from FAOSTAT Agricultural Producer Prices.

The economic extension uses annual Producer Price (USD/tonne) values for cassava, taro and sweet potato.

Complete same-year USD producer prices for all three crops were available for 10 selected stress events:

Fiji:

1998, 2003, 2004, 2006, 2015

New Caledonia:

2015, 2016, 2017, 2018, 2019

For each strategy, expected and historical crop quantities are valued using that country-crop-year's observed producer price."

Then state prominently:

"The resulting quantity is gross farm-gate value relative to the fitted harvest benchmark. It is not farmer profit, household income or retail food value."

Also:

"Observed same-year producer prices are used as valuation weights. This analysis does not estimate how drought itself changes market prices."

DOWNSIDE DEFINITION

Add:

"Economic downside is defined as the negative portion of the farm-gate-value outcome:

downside = $\max(-\text{value outcome} \%, 0)$."

SINGLE-CROP BENCHMARK

Add:

"The no-diversification benchmark is the equal-probability average downside from committing the full hectare to one of cassava, taro or sweet potato before the outcome is known.

This is a hypothetical ex-ante risk benchmark. It is not an estimate of actual farmer monocropping behaviour."

STATISTICAL SUPPORT

Add:

"Across the 10 price-covered stress events, the average single-crop benchmark downside is approximately 6.63%, compared with 3.10% for the equal-third diversified hectare — a 3.53 percentage-point difference, or approximately 53.3% proportional reduction in average downside."

Then:

"Supporting paired checks give an exact sign-flip p-value of about 0.0078 and a paired Wilcoxon p-value of about 0.0122. An event-level bootstrap gives an approximate 95% interval of 1.37–5.91 percentage points."

Then the essential caveat:

"These inferential checks are exploratory. The price-covered sample contains only 10 events clustered within Fiji and New Caledonia, so the effect size and observed event pattern should carry more weight than a Pacific-wide interpretation of the p-values."

SOURCES

Add FAOSTAT Agricultural Producer Prices to the source list.

Dataset catalogue:

<https://data.fao.org/catalog/dataset/47c17894-8ca1-4bd0-ba4d-6078e919e9b1>

Retain FAOSTAT QCL for crop production / area.

Retain Pacific Data Hub climate indicators.

Do not remove the current ABC, Nunn et al., Kaoh et al., ACIAR, GCF, SPC/CePaCT, FAO and IPCC references unless one is genuinely unused.

PROMPT 12 - Final polish and QA

Final QA pass for Roots of Resilience.

Do not redesign anything during this stage.

Only fix errors, inconsistencies, responsiveness and polish.

STORY FLOW

Verify the progress rail follows exactly:

hero
warming
plant
tonga
sixty
winner
joint
downside
money
insurance
islands
resilience
meaning
close

Verify the sticky visual displayed for every beat matches its data-visual attribute.

Verify SCENE_FOR and VISUAL_TITLES contain every new beat.

There must be no obsolete references to:

safe
fail
why
evidence

inside BEAT_IDS or the main sticky visual navigation.

Old functions are allowed only if they are still deliberately used by Explore the model.

COPY SEARCH

Search the full source and remove/replace any stale phrase:

"All three failed"
"all three roots failed"
"every crop failed"
"the same principle this dataset measures another way"
"taro alone carries ceremonial weight across the Pacific"

Search for any claim that says diversification:

always wins
always performs better
prevents crop failure
is causal
is the optimal Pacific planting mix

None should remain.

NUMERICAL QA

Confirm final site displays:

60 stress events
10 per country

Best counts:
SP 25
Taro 21
Cassava 14

Joint:
56 / 60 at least one at/above fitted trend
4 / 60 all three below fitted trend
6.67%, not 400%

Fiji 2004:
Cassava +23.5%, +\$920
Taro -14.0%, -\$968
Sweet potato -35.3%, -\$680
Mixed -5.7%, -\$243

Economic:
6.6% -> 3.1%
53.3% lower average downside
10 price-covered events
Fiji + New Caledonia

Do not accidentally label the 53.3% result as:
"53.3% lower dollar losses"

It is:
"53.3% lower average percentage downside in farm-gate value."

If any hot+dry count is displayed anywhere:
the current analysis contains 6 hot+dry stress events, not 4.

RESPONSIVENESS

Test at approximately:

1440px desktop
1024px laptop/tablet landscape
768px tablet
390px mobile
360px small mobile

On mobile:

- sticky visual must not cover the narrative text
- heatmap must remain interpretable
- no tooltip may leave viewport
- tap must replace hover
- no text under 10–11px for meaningful information
- buttons/tap targets minimum ~44px
- charts may horizontally scroll if necessary
- never compress a dense chart until labels become unreadable

ACCESSIBILITY

Keep prefers-reduced-motion support.

All interactive SVG elements should be keyboard reachable or have an accessible equivalent.

Do not communicate positive/negative outcomes using colour alone.

Use:
colour + direction + symbol/text.

All charts need useful aria-label text.

Tooltips must not be the only place essential claims are stated.

Theme switching must preserve sufficient contrast in both:

Ocean Garden
Field Notes

PERFORMANCE

Do not load the six new visuals repeatedly during scrolling.

Load data once.

Render static visual geometry once where possible.

Use class changes / opacity / attributes for interaction instead of rebuilding an entire SVG on every mousemove.

Do not run expensive portfolio calculations on scroll.

FINAL STORY TEST

Ask a person unfamiliar with the project to scroll through once.

Without opening Methods, they should be able to answer:

1. What decision did I make?
2. Why can't I simply pick the "best" crop?
3. What does diversification change?
4. Why do dollars matter?
5. Did diversification win every year?
6. How limited is the economic evidence?
7. Why isn't one crop mix a universal Pacific prescription?

If any of those are unclear, improve transitions rather than add more statistics.

Implementation priority

Non-negotiable interaction #1: On the 60-event downside graph, hovering or tapping one country-year should highlight that same event across Cassava, Taro, Sweet potato, and Diversified. This makes the core risk-diversification logic visible without requiring a paragraph of explanation.

Priority #2: On the winner heatmap, hovering a crop should trace that crop across all six countries so the viewer can visually watch its performance change through time and place.

Priority #3: On the insurance chart, preserve and visibly show the red exceptions where diversification increased downside. The story is stronger because the chart does not pretend diversification automatically wins every year.

Final principle: Focus on story choreography rather than adding more statistics. The existing site already has a strong foundation; the next gain comes from making the quantitative evidence unfold coherently and interactively.